HW 5 Cost Curves

Q1 $w = 1$ fixed cost of $FC = 2$.

$MP_L \Rightarrow$ measures the additional output from hiring one more workes.

$MP_L = Q_L - Q_{L-1}$ $\quad VC = w \cdot L$

$AP_L = \frac{Q_L}{L}$ $\quad AVC = \frac{L}{Q}$ $\quad AFC = \frac{2}{Q}$

$MC = \frac{\Delta TC}{\Delta Q} = \frac{1}{MP_L}$ $\quad TC = L + 2$ $\quad ATC = \frac{L+2}{Q}$

Labor	Output Q	MP	AP	Wage	VC	AVC	FC	TC	ATC	MC
0	0	—	—	1	0	—	2	2	—	—
1	6	6	6	1	1	$\frac{1}{6}$	2	3	$\frac{1}{2}$	$\frac{1}{6}$
2	15	9	$\frac{15}{2}$	1	2	$\frac{2}{15}$	2	4	$\frac{4}{15}$	$\frac{1}{9}$
3	27	12	9	1	3	$\frac{1}{9}$	2	5	$\frac{5}{27}$	$\frac{1}{12}$
4	37	10	$\frac{37}{4}$	1	4	$\frac{4}{37}$	2	6	$\frac{6}{37}$	$\frac{1}{10}$
5	45	8	9	1	5	$\frac{1}{9}$	2	7	$\frac{7}{25}$	$\frac{1}{8}$
6	50	5	$\frac{25}{3}$	1	6	$\frac{3}{25}$	2	8	$\frac{4}{25}$	$\frac{1}{5}$
7	52	2	$\frac{52}{7}$	1	7	$\frac{7}{52}$	2	9	$\frac{9}{52}$	$\frac{1}{2}$
8	50	-2	$\frac{25}{4}$	1	8	$\frac{4}{25}$	2	10	$\frac{1}{5}$	$-\frac{1}{2}$

AP $L = 8$ the table produces $MC = -\frac{1}{2}$ because total cost rises by 1 while output falls by 2.

uestion 2 : Total cost curves

VC ; TFC ; TC relationship = TC = TFC + TVC

FC : Always 2 ⇒ it's horizontal line

VC : Equals labor because the wage is 1.

TC : Equals TVC plus 2.

Output Q	TVC	TFC	TC
0	0	2	2
6	1	2	3
15	2	2	4
27	3	2	5
37	4	2	6
45	5	2	7
50	6	2	8
52	7	2	9
50	8	2	10

The last row

returns to output 50 because the eigth worker reduces production from 52 to 50. Horizontal coordinate is output,

Question 3: Figure shows short-Run total cost when q = 15

Wage per worker = 1 VC = 2x1 = 2 FC = 2

Therefore STC = TFC + TVC = 2+2 = 4

$STC = ATC \times q = \frac{4}{15}$ $STC = \frac{4}{15} \times 15 = 4$

The ATC curve is approx. 0.267 at q = 15

Question 4 TFC when $q = 27$

At output $q = 27$, the firm employs three workers. FC is given as 2. does not change with output.

$$TFC = 2$$

$$AFC = ATC - AVC$$

at $q = 27$ the graph gives approx. $AFC \approx 0.185 - 0.111 = 0.074$

$$TFC = AFC \times q \approx 0.074 \times 27 \approx 2$$

Ans: $TFC = 2$

The vertical distance between the ATC & AVC curves represents AFC. Multiply that difference by output to recover total fixed cost.

Question 5 show TVC when $q = 27$

$q = 27$ wage $= 1$ $TVC = w \times L = 1 \times 3 = 3$

Average variable cost curve $TVC = AVC \times q$.

at $q = 27$ $AVC = \frac{1}{9}$ then $TVC = \frac{1}{9} \times 27 = 3$

Answer TVC : 3